

HOMEWORK 2

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Problem 1. Let $b > 1$.

- (a) If m, n, p, q are integers, $n > 0, q > 0$, and $r = \frac{m}{n} = \frac{p}{q}$, prove that

$$(b^m)^{1/n} = (b^p)^{1/q}.$$

Hence it makes sense to define $b^r = (b^m)^{1/n}$.

- (b) Prove that $b^{r+s} = b^r b^s$ if r and s are rational.

- (c) If $x \in \mathbb{R}$, define

$$B(x) = \{b^t : t \in \mathbb{Q}, t \leq x\}.$$

Prove that $b^r = \sup B(r)$ when $r \in \mathbb{Q}$, and use this to define

$$b^x = \sup B(x) \text{ for every } x \in \mathbb{R}.$$

- (d) Prove that $b^{x+y} = b^x b^y$ for all real x, y .

Proof. A. We are given that $\frac{m}{n} = \frac{p}{q} = r$, and an arbitrary real number $b > 1$. We can clearly deduce that $b = b$. Now raise b to the power of r for both sides: $b^r = b^r$. Now, for the LHS, let $r = \frac{m}{n}$, and $r = \frac{p}{q}$ for the RHS: $b^{m/n} = b^{p/q}$. Now, through exponent rules, we arrive at the conclusion that $b^r = (b^m)^{1/n} = (b^p)^{1/q}$. $\square$

Proof. B. We are given that r and s are rational numbers, and are asked to prove the statement $b^{r+s} = b^r b^s$. Let $r = \frac{m}{n}$ and $s = \frac{p}{q}$, where $m, n, p, q \in \mathbb{Z}, n, q \neq 0$. We can use this to show that $r + s = \frac{mq+np}{nq}$. Now, raise b^{r+s} , which may be denoted as $b^{\frac{mq+np}{nq}}$. We can deduce that $mq + np$ is addition between two integers. So, let $\alpha = mq$ and $\beta = np$, where $\alpha, \beta \in \mathbb{Z}$. We must show that $b^{\alpha+\beta} = b^\alpha b^\beta$. Assuming $\alpha, \beta \geq 0$, we can illustrate their equivalence:

$$\underbrace{b \cdot \dots \cdot b}_{\alpha+\beta \quad b's} = \underbrace{(b \cdot \dots \cdot b)_{\alpha+\beta} (b \cdot \dots \cdot b)_{\beta}}_{\alpha+\beta \quad b's}$$

therefore, $b^{\alpha+\beta} = b^\alpha b^\beta$. If either α or $\beta < 0$, then we can represent this as $b^{-\alpha} = \frac{1}{b^\alpha}$.

We can use this result now to show that $(b^\alpha b^\beta)^{\frac{1}{nq}}$. We can do this because of our result in **A**. Now, we can distribute the power into the b 's, giving us $b^r b^s$. Thus, $b^{r+s} = b^r b^s$. $\square$

Proof. C. We are given the set $B(x) = \{b^t : t \in \mathbb{Q}, t \leq x\}$, and we are trying to prove when our input $r \in \mathbb{Q}$, then $b^r = \sup B(x)$. Consider a rational r such that $B(r) = \{b^t : t \in \mathbb{Q}, t \leq r\}$. Now we have the inequality $t \leq r \implies b^t \leq b^r$, where $b^r \in B(r)$ due to r 's rationality. We can show that this is the case due to the monotonicity of the set. Since both $t, r \in \mathbb{Q}$, then $\exists! t = r$. Thus, since $t \leq r$, $\max B(r) = b^r \implies \sup B(r) = b^r$. Now consider the real number x such that $B(x) = \{b^t : t \in \mathbb{Q}, t \leq x\}$. We can show that $B(x)$ is non empty since we know

that there exists a rational $-r < x < r$, so $B(-r)$ exists, and $B(x) \neq \emptyset$. Now we can choose a rational r that is greater than x and we show that $x < r \implies b^r$ is an upper bound of $B(x)$. Therefore, by the L.U.B property, $\sup B(x)$ exists. If x is rational, then $\sup B(x) = b^x$ by definition. This applies to $x \in \mathbb{R}$. Thus, $\sup B(x) = b^x$. $\square$

Proof. D. Through **B**, we've shown that if x, y are both rationals, then this works out. Therefore, let's assume that x or y are irrational. Using the result from **C**, we can take two rationals $r + s \leq x + y$, where $r \leq x$ and $s \leq y$, which gives $b^{r+s} \in B(x+y)$, and, using **B**, $b^{r+s} = b^r b^s$. Since $b^r \leq b^x$ and $b^s \leq b^y$ so $b^r b^s \leq b^x b^y$. Thus, $b^{x+y} \leq b^x b^y$. We can also show that $b^{r+s} \leq b^{x+y} \implies b^x b^y \leq b^{x+y}$. So, $b^{x+y} = b^x b^y$. $\square$

Problem 2. If $z \in \mathbb{C}$ satisfies $|z| = 1$ (that is, $z\bar{z} = 1$), compute

$$|1 + z|^2 + |1 - z|^2.$$

Proof. We know $z = x + iy$ and $\bar{z} = x - iy$ for $x, y \in \mathbb{R}$, and that $|z| = \sqrt{x^2 + y^2}$. Now, we have $|1 + x + iy|^2 + |1 - x - iy|^2 = (1 + x)^2 + y^2 + (1 - x)^2 + y^2 = 2 + 2x^2 + 2y^2 = 2(1 + x^2 + y^2) = 2(1 + |z|^2)$. $\square$

Problem 3. Suppose $k \geq 3, x, y \in \mathbb{R}^k, |x - y| = d > 0$, and $r > 0$. Prove:

- (a) If $2r > d$, there are infinitely many $z \in \mathbb{R}^k$ such that $|z - x| = |z - y| = r$.
- (b) If $2r = d$, there is exactly one such z .
- (c) If $2r < d$, there is no such z .

How must these statement be modified if $k = 2$ or $k = 1$?

Proof. Let $d = d(x, y)$, and let $m = (x + y)/2$. Let z be perpendicular to the line segment generated by x, y . We see that $d(z, m) = \sqrt{r^2 - (d/2)^2} = R$. From this, we see that $d(z, m) > 0$ and it tells us the distance between m and z . Denote this as L . From this, we see that $r = d/2 \implies L = \sqrt{r^2 - (d/2)^2} > 0$. Since $k \geq 3$ and $L > 0$, this actually forms a sphere, thus making z be infinitely many points as long as it is perpendicular and equal to L . If $r = d/2$, then $d(z, m)^2 = r^2 - d^2/4 = 0$, so $z = m$, proving that there is only one z that works. If $r < d/2$, then $r^2 - d^2/4 < 0$, but $d(z, m) > 0$, so this is impossible. If $k = 2$, then you wouldn't get infinite points for the first case, and only get one because it is no longer a sphere. For $k = 1$, There would be no solutions because you're practically on a line. $\square$